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Soil water repellency in a Japanese cypress plantation restricts increases in soil water storage during rainfall events

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Abstract:

Forest soils in Japan are often water repellent. Substantial water repellency frequently occurs and impedes water infiltration into the soil matrix, but continuous overland flow is not necessarily observed because forest soils usually have macropores through which the water can enter the subsoil. Although this flow pattern may influence the manner of water storage in forest soils at the solum scale, field evidence has not yet indicated this process. We monitored soil water storage during natural rainfall events in a 60-cm deep solum using time domain reflectometry (TDR) moisture sensors, and observed stained flow patterns in the soil following simulated rainfall containing a colour dye, on a slope planted with Japanese cypress (*Chamaecyparis obtusa*). The surface soil at the research plot exhibited strong water repellency at water contents lower than the threshold critical water content of 0·29 m^3 m^{-3}. Under dry antecedent moisture conditions, increases in soil water storage were small compared to the cumulative rainfall, despite the low wetness of the soil matrix. In contrast, under moderate moisture conditions, increases in the water content corresponded to the cumulative rainfall. Under dry conditions, rainwater may have entered the subsoil at a few limited locations connected with continuous vertical macropores, such as decayed root channels or interstructural voids. Therefore, the water seemed to bypass a large part of the soil matrix away from the macropores. Such preferential water flow was confirmed by the stained flow patterns after the rainfall simulation. The flow patterns visualized by the dye were discontinuous and scattered under dry conditions and diffuse under moderate moisture conditions. Repellency induced preferential flow led to restricted increases in solum scale water storage during rainfall events, reflecting a physical nonequilibrium in soil water storage.

KEY WORDS forest soil; macropores; physical nonequilibrium; preferential flow; soil water storage; water repellency

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INTRODUCTION

Water repellency is an important soil physicochemical property that strongly affects near surface soil water transport by creating a reduced infiltration capacity (Bond, 1964; Murai and Iwasaki, 1975; Wallis *et al.*, 1991) and resultant overland flow (Krammes and DeBano, 1965; McGhie, 1980), fingerlike flows in structurally homogeneous soils (Raats, 1973; Dekker and Jungerius, 1990), and bypass flows in heterogeneous soils (Burch *et al.*, 1989; Imeson *et al.*, 1992; Kobayashi, 2002). Water repellent soils are distributed worldwide (Dekker *et al.*, 2005), and soil water repellency has been reported in Japan (Nakaya, 1982; Miyata *et al.*, 2007), which is characterized by a warm–humid monsoon climate. Some soils in dry environments have long been recognized as water repellent, particularly in forests, which cover 67% of the land area of Japan (Ohmasa, 1951).

In forested mountainous regions, strong soil water repellency caused by wildfires and resultant overland flow often leads to serious erosional damage (Wells *et al.*, 1979; Shakesby *et al.*, 2000). Studies have also described unburned forest soils that exhibit water repellency as strong as that of burned soils (Reeder and Jurgensen, 1979; Doerr *et al.*, 1998). However, as pointed out by Doerr *et al.* (2000) and Shakesby *et al.* (2000), continuous overland flow over entire slopes does not seem to occur on unburned forested slopes because of water capture by sinks or gaps, e.g. hydrophilic patches, cracks, and holes. In the case of forest soils, continuous macropores, which are commonly found in the near surface layer, are thought to allow repellency induced overland flow to enter the subsoil (Burch *et al.*, 1989; Willson *et al.*, 1990; Doerr *et al.*, 2000). If such a process is dominant in a soil, rainwater would move downward as preferential flow through macropores without spreading into the soil matrix, at least at depths where the soil is water repellent.

Several studies have examined preferential flow caused by water repellency from the perspective of the environmental risks caused by the rapid transport of agricultural chemicals or pollutants to the groundwater (e.g. Hendrickx *et al.*, 1993; Ritsema *et al.*, 1993). However, for water conservation in mountainous regions, it is also necessary to evaluate the effect of soil water repellency on the amount of water stored in soils. Soil water repellency may affect the soil water storage that controls water flow regulation in a catchment. Few studies, however, with

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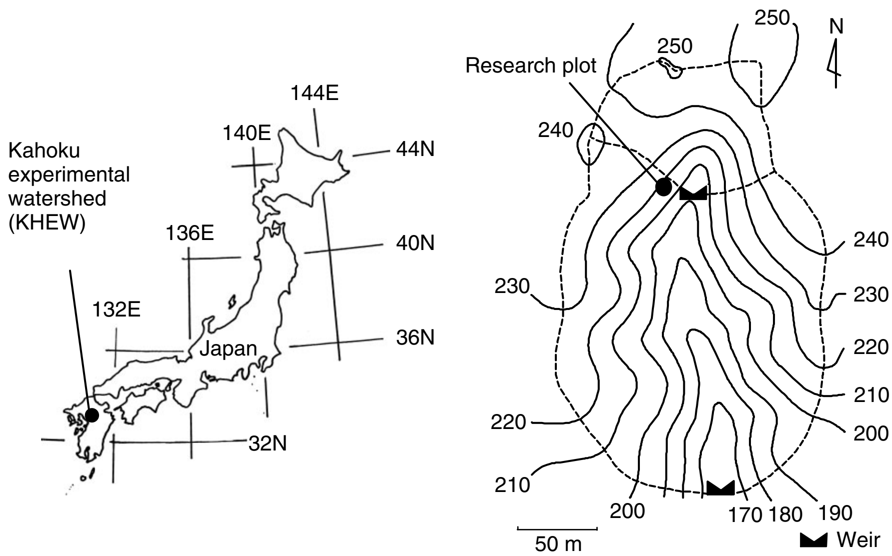

Figure 1. Location and topography of the studied watershed. The contour interval is in metres

the exception of work by Dekker and Ritsema (1996) on water absorption into an undisturbed column of water repellent grassland soil, have attempted to explore the effect of water repellency on soil water storage. In particular, water storage in water repellent soils under field conditions has not been demonstrated.

We sought to clarify the effect of water repellency on solum scale water storage in forest soils by monitoring soil water storage during natural rainfall events using time domain reflectometry (TDR) and conducting rainfall simulations using a colour dye on a forested slope in a small headwater catchment.

MATERIALS AND METHODS

Study area

The study was conducted at the Kahoku experimental watershed (KHEW) of the Forestry and Forest Products Research Institute in the Nagaso National Forest located in northern Yamaga, Kumamoto Prefecture, Japan (33·13 °N, 130·07 °E; Figure 1). Detailed information about the study area has been provided by Shimizu *et al*. (2003). The mean annual precipitation from 1992 to 2003 was 2160 mm, and the mean annual air temperature was 15·4 °C. Crystalline schist underlies the watershed.

The research plot was established in the centre part of a sideslope in the watershed (Figure 2); the slope angle around the research plot was 35°. This watershed is used as a coniferous tree plantation. Japanese cedars (*Cryptomeria japonica*) are planted on the lower slopes of the catchment, and Japanese cypress (*Chamaecyparis obtusa*) occupies the middle and upper slopes. In some locations, planted conifers had been suppressed by broad leaved trees such as *Quercus glauca, Quercus serrata*, and *Castanopsis cuspidata*. Around the research plot, Japanese cypress exclusively formed the canopy. The average height and the average diameter at breast height of the cypress trees were 17·9 and 13·8 cm, respectively.

Figure 2. Vegetative condition around the research plot

Ferns (*Gleichenia japonica*) densely covered the forest floor around the research plot.

The soil was brown forest soil (Cambisol by the FAO classification), and the texture of the fine earth was clay loam. The gravel content was 6% by volume (diameter range: 2–10 mm). The bulk density was 1·02 mg m^{-3}. Deeper soil (>70 cm in depth) contained stones >50 mm in diameter. The soil at depths of 0 to 50 cm had a weak, coarse, sub-angular, blocky structure. Some visible gaps (0·5–1 mm) around living roots and stones were observed. A litter layer composed of leaves from the cypress trees and the ferns ranged in thickness from 0·5 to 1 cm. The leaf litter fragments were small, especially those of cypress litter, which easily breaks into fragments <3 mm. Therefore, the litter layer seemed to have little effect as a barrier to water entry.

Water repellency of the soil in the observation plot

The soil in the watershed, excluding that near the stream, exhibited strong water repellency, which occurs under dry conditions and disappears under wet conditions (DeBano, 1971). We examined the relationship between water repellency and soil moisture for the surface soil at depths of 0–5 cm in the research plot as follows.

Field moist soil was sieved using a 2-mm mesh screen and dried at room temperature for 3 weeks. The air dried soil was rewetted to standardize the wetting history and dried again at room temperature for 30 min to 24 h. Using this method, we prepared 30 samples of different moisture content. The degree of water repellency was measured using the ethanol percentage (EP) as an index (Watson and Letey, 1970). Aqueous ethanol solutions with different concentrations of 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 18, 20, 22, 24, 27, 30, 35, 40, 45, 50% by volume were prepared. The EP was defined as the concentration when a droplet of the solution penetrated into a smoothed soil sample in 5 s. The gravimetric water content (kg kg^{-1}) of each sample was determined by oven drying (105 °C) and transformed into volumetric water content (m^3 m^{-3}) using the bulk density of the soil.

Rainfall simulations using a dye tracer

To observe the flow pattern in the soil, we conducted rainfall simulations on the slope using a dye tracer. We applied a depth equivalent of 30 mm of water containing a colour dye (Brilliant Blue FCF; Flury and Flühler, 1995) to the soil using a rainfall simulator at a rate of 20 mm h^{-1}. The concentration of the dye was 2·5 g L^{-1}. The simulator had 240 needles arranged uniformly in a 90 × 90 cm area. Water was supplied at a constant rate from a Mariotte tank (constant head device) from hight of 80 cm in average. To spread the 'raindrops' homogeneously over the 1 × 1 m application area, a nylon screen with 1 mm mesh was placed between the needles and the soil surface. Rainfall simulations were conducted under a moderate moisture condition and dry moisture condition; the initial water contents at the 10 cm depth at the onset of rainfall (θ_i) were 0·34 and 0·25 m^3 m^{-3}, respectively. After water applications, the soil was immediately excavated, and the flow patterns indicated by the dye on the soil profiles were examined.

Measurement of water content and calculation of soil water storage during natural rainfall events

Soil water content was monitored at depths of 10, 30, and 50 cm using TDR to determine the change in solum scale soil water storage during natural rainfall events. The probes of the TDR sensor (CS615; Campbell Scientific, Inc., Logan, UT, USA) were 30 cm long. A single sensor was installed at each depth. One sensor is thought to measure the average water content of a cylindrical volume with length of 30 cm and diameter of approximately 8 cm. All sensors were inserted into the soil horizontally from the sidewall of a soil pit. The pit was backfilled after the installation. Measured water content values were recorded every hour with a datalogger. The analyzed water content data in this study were measured from September 2001 to December 2002. Thirty-seven events with total rainfall greater than 20 mm were observed in this period.

Solum scale soil water storage was defined and calculated as follows. Water contents measured at depths of 10, 30, and 50 cm (θ_{10}, θ_{30}, and θ_{50}) were considered as representative for 0–20, 20–40, and 40–60 cm intervals, respectively. The solum scale water storage at depths of 0–60 cm (S: mm) was calculated as $S = \Delta z(\theta_{10} + \theta_{30} + \theta_{50})$, where Δz is the thickness of the intervals (200 mm). The increase in the water storage (ΔS) for an arbitrary time during a rainfall event was determined by $\Delta S = S - S_i$ (S_i = water storage at the onset of rainfall). For rainfall events with a total amount exceeding 20 mm, ΔS at the moment when cumulative rainfall reached 20 mm was calculated and defined as ΔS_{20}. Ideally, ΔS_{20} should be 20 mm if all rainwater is stored in the upper 60 cm of the soil, assuming no interception, and 0 mm if none of the rainfall is stored. While the canopy interception rate varies in each rainfall event, we considered the variation to have been small when the total rainfall was over 20 mm; therefore, we used net rainfall as the index in this study. For larger events, ΔS_{30} and ΔS_{40} were calculated in the same manner. In this paper, ΔS_{20}, ΔS_{30}, and ΔS_{40} are used as indices of the solum scale water storage.

Numerical simulations of soil water storage assuming no water repellency

Numerical simulations of the change in soil water storage were conducted using the computer program HYDRUS-2D (Šimůnek *et al.*, 1999) assuming no soil water repellency. The hydraulic parameters were experimentally obtained using an undisturbed cylindrical soil sample, which was taken near the measurement location. The sampling depth was 30 cm. The sample exhibited no water repellency. The relationship of the water content and matric potential was measured by the pressure plate method. The hydraulic conductivity–matric potential relationship was determined by the flux-head steady state measurment. The obtained relationships were represented by the soil hydraulic functions of van Genuchten (1980). The model parameters are listed in Table I. One dimensional water movement in uniform soil of 100 cm thickness was simulated using constant flux as the upper boundary condition and free drainage as the lower boundary condition, assuming no water repellency, no interception, and no evapotranspiration. Then the values of ΔS_{20}

Table I. The model parameters of soil-hydraulic functions of van Genuchten (1980)

θ_s	θ_r	α	n	K_s	l
0.259	0·524	0·106	1·781	$2{\cdot}7 \times 10^{-4}$	0·67

Unit of hydraulic conductivity K_s: m s^{-1}

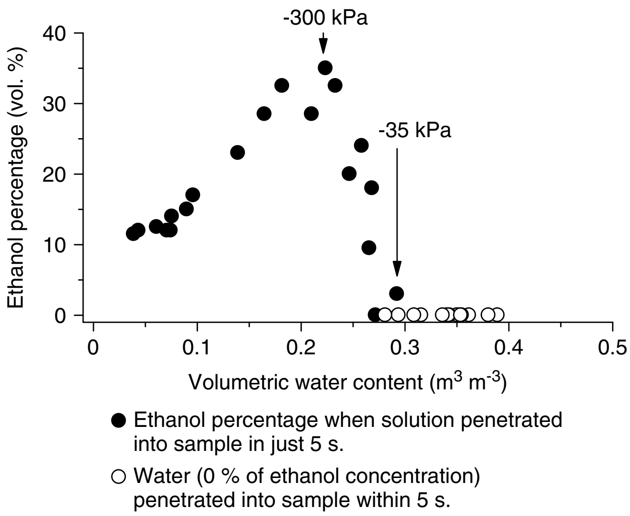

Figure 3. Relationship between water content and degree of water repellency measured by the ethanol percentage (EP) index for soil depths of 0 to 5 cm in the research plot. The matric potential values correspond to the critical water content and the water content for the area of highest measured water repellency

for uniform initial water content ranging from 0·15 to 0·45 $m^3\ m^{-3}$, under constant rainfall intensities of 1, 5, and 20 mm h^{-1}, were calculated.

RESULTS

Relationship between soil moisture and water repellency

Water repellency of the soil in the research plot was observed when the water content of the soil was below 0·29 $m^3\ m^{-3}$ (Figure 3). Above this water content, water repellency did not occur. This value can thus be regarded as the 'critical water content' (Dekker and Ritsema, 1994) for water repellency of this soil. The corresponding matric potential for the critical water content was −35 kPa. The degree of water repellency was the highest at a water content of approximately 0·22 $m^3\ m^{-3}$, which is equivalent to a matric potential of −300 kPa and actually measured in the field. When the soil became drier (dry conditions occurred only in the laboratory), the degree of water repellency decreased.

Observed flow patterns in the soil under different antecedent moisture conditions after the rainfall simulations

Under a moderate antecedent moisture condition ($\theta_i = 0{\cdot}34 >$ critical water content), no overland flow was observed during the application of simulated rainfall, i.e. water infiltrated into the soil within the application area. The stained area on the excavated soil profile was essentially continuous from the soil surface to the subsoil, although it seemed somewhat irregular (Figure 4a). Because of the adsorption, the movement of the dye should have been retarded in the soil (Flury and Flühler, 1995). The actual flow pattern of the rainwater, therefore, should have been more diffuse than that indicated by the dye. Accordingly, it was thought that the majority of the soil matrix participated in water transport and storage.

In contrast, under a dry antecedent moisture condition ($\theta_i = 0{\cdot}25 <$ critical water content), part of the water supplied from the rainfall simulator formed overland flow and runoff outside the water application area. The flow pattern on the profile was remarkably discontinuous and scattered (Figure 4b). The total stained area was very small compared to the entire soil profile. Particularly at depths of 0–30 cm, the soil matrix seemed to retain little of the applied water. Most of the stained areas converged along specific structures such as holes formed by decayed roots, gaps between the soil and living roots, and interstructural voids.

Measured water content under natural rainfall

Figure 5 shows changes in soil water content during natural rainfall events for moderate and dry moisture conditions. The propagation patterns of water content under dry conditions differed from patterns formed in moderate moisture conditions. Figure 5a shows an example for a moderate moisture condition ($\theta_i = 0{\cdot}35\ m^3\ m^{-3}$). There had been 29 mm of rain in the week before this period. Therefore, the water content at the soil surface was thought to have been similar to that at 10 cm and higher than the critical water content for water repellency ($= 0{\cdot}29\ m^3\ m^{-3}$). In this example, the water contents at

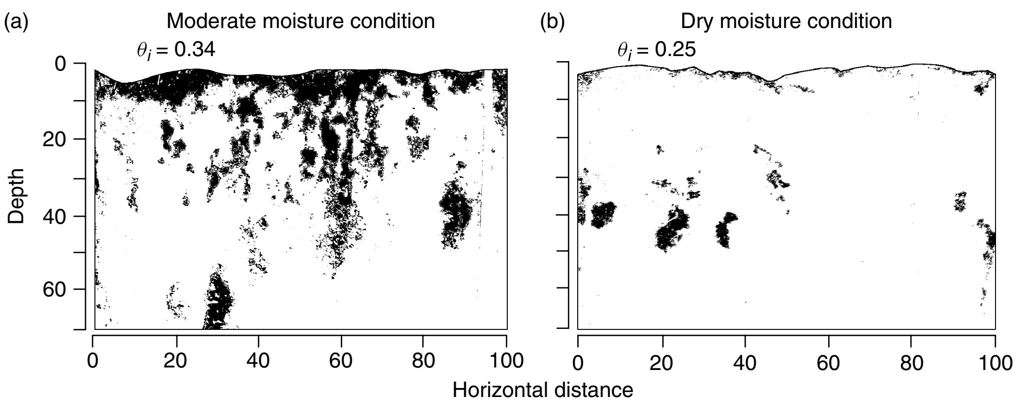

Figure 4. Flow patterns revealed by the colour dye during rainfall simulations conducted under (a) moderate and (b) dry antecedent moisture conditions

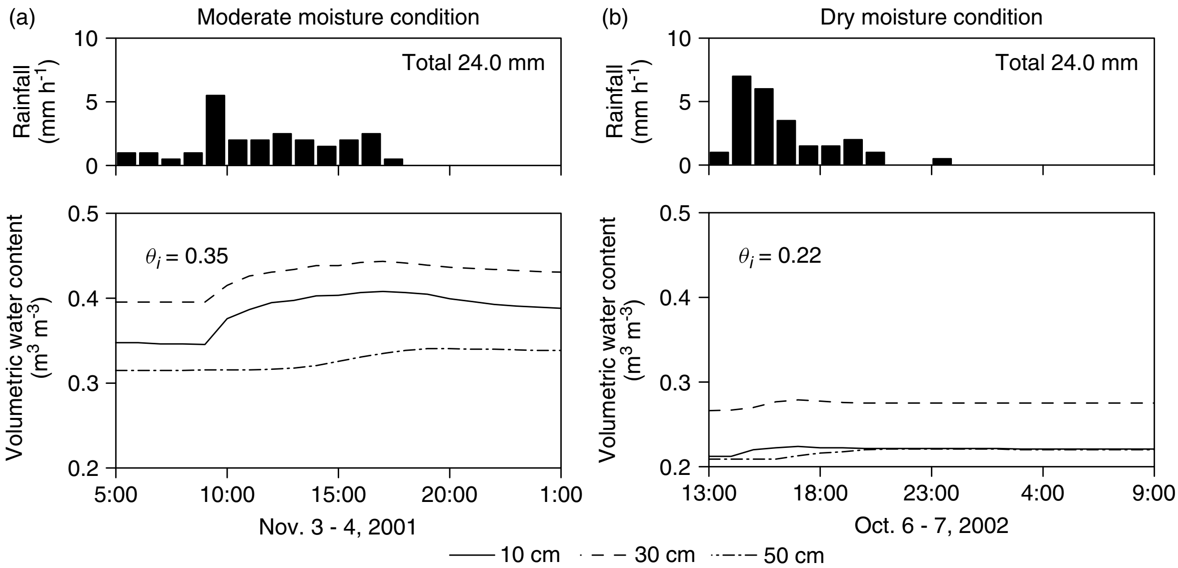

Figure 5. Typical changes in water content measured using TDR moisture sensors during natural rainfall events under (a) moderate and (b) dry antecedent moisture conditions

depths of 10 and 30 cm initially increased by approximately 0·05 m^3 m^{-3}. Five hours later, the water content at 50 cm increased.

In contrast, under the dry moisture condition (Figure 5b), θ_i was 0·22 m^3 m^{-3}; accordingly, the water content of the soil surface was presumed to have been lower than the critical water content. Although the water contents at 10 and 30 cm increased by only 0·01 m^3 m^{-3}, a change in water content propagated to the depth of 50 cm in 2 h. The tendency for quick propagation without a major increase in the water content under dry conditions is contrary to the expected behavior of so-called normal soils.

Increases in soil water storage during natural rainfall events

The relationship between cumulative rainfall and water storage increases varied considerably with antecedent moisture conditions. Under moderate conditions, we found parallel increases in cumulative rainfall and ΔS (Figure 6a). This result indicates that the rainwater was almost fully stored within the top 60 cm of the soil. On the other hand, under dry conditions, ΔS was small compared to the cumulative rainfall (Figure 6b), i.e. the ability of the soil to store water at the solum scale seemed to have been reduced. Furthermore, in this example, the increase of ΔS seemed to peak at 5 mm, although considerable rainfall continued. This suggests that the water holding capacity of the soil was temporarily reduced to only 5 mm.

Relationship between antecedent moisture conditions and water storage

Figure 7 shows the relationship between antecedent moisture conditions and changes in water storage during natural rainfall events. Numerically simulated values of

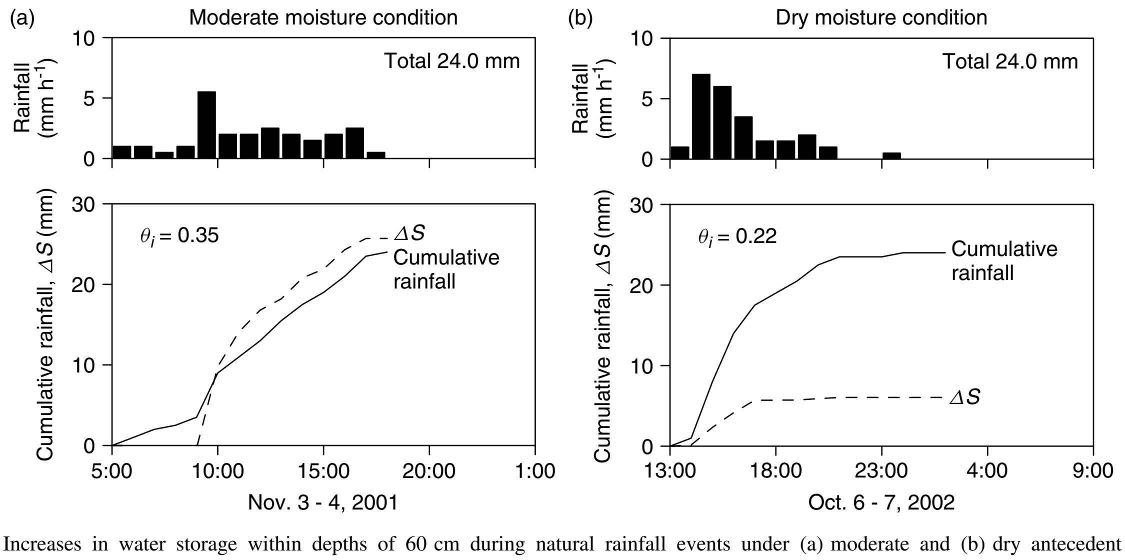

Figure 6. Increases in water storage within depths of 60 cm during natural rainfall events under (a) moderate and (b) dry antecedent moisture conditions

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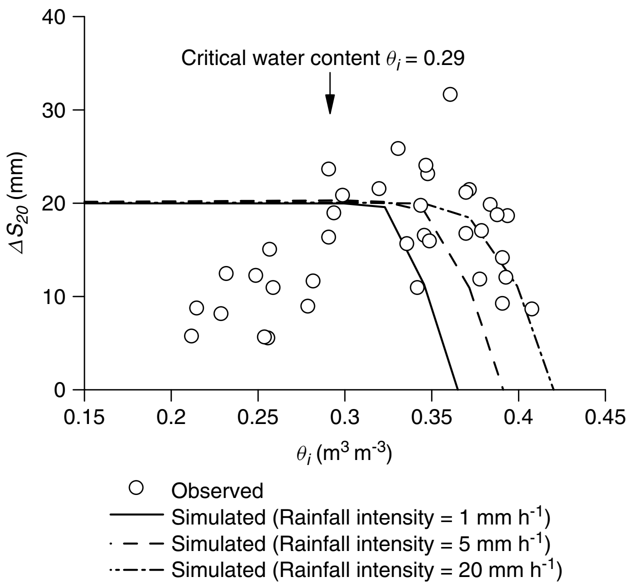

Figure 7. Relationship between the initial water content at a depth of 10 cm (θ_i) and increases in water storage within a depth of 60 cm during rainfall events for cumulative rainfalls of 20 mm (ΔS_{20})

ΔS_{20} for constant rainfalls with various intensities are also indicated in this figure.

In cases in which the initial water content θ_i was higher than the critical water content of 0·29 m^3 m^{-3} (i.e., moderate or wet antecedent moisture conditions), the observed and simulated ΔS_{20} agreed well. Decreases in ΔS_{20} occurred with increases in θ_i. This tendency should be common for homogeneous wettable soils because a soil with high water content has little capacity for additional water storage. In contrast, in cases in which θ_i was lower than the critical water content (i.e. dry antecedent moisture conditions), the observed ΔS_{20} decreased with decreases in θ_i, while the simulated ΔS_{20} remained constant (= 20 mm). The largest discrepancy between observed and simulated ΔS_{20} was approximately 15 mm. The tendency of decreasing ΔS with decreasing antecedent wetness was clear, although the data varied considerably and sometimes exceeded the input rainfall of 20 mm under moderate and wet antecedent conditions. The same trend was observed in the cases for cumulative rainfalls of 30 and 40 mm (Figure 8).

DISCUSSION

Preferential water flow caused by water repellency

Under dry antecedent moisture conditions, water content change propagated to the deeper soil without major water content increases in the upper soil (Figure 5b). This indicates that rainwater flowed to the deeper soil only through spatially limited flow paths. The discontinuous flow paths stained by the dye (Figure 4b) indicated preferential water movement in the soil. Moderate antecedent moisture conditions brought a contrasting result. The water content of the deeper soil increased after a substantial increase in water content of the upper soil (Figure 5a). The staining pattern in the soil profile was more diffuse under moderate moisture conditions (Figure 4a) than under dry conditions. The homogeneity of water flow under moderate moisture conditions was higher than under dry conditions. The soil examined in this study had a weak sub-angular blocky structure, and contained macropores between structural units or around living roots and stones. And the soil exhibited strong water repellency under dry conditions. In the rainfall simulation under the dry condition (i.e., the initial water content of the soil surface was lower than the critical water content for repellency), water was not absorbed immediately into the soil matrix, and part of the water ran off as overland flow. These results indicate that the observed vertical water movement under dry conditions was "bypass flow" through continuous macropores open to the soil surface. Field observations have suggested the occurrence of vertical preferential flow caused by repellency (Burch *et al.*, 1989; Willson *et al.*, 1990). We confirmed this tendency through observations and measurements. Preferential flow along living roots and through decayed root channels in an unsaturated soil has also been reported by Noguchi *et al.* (1999). Water repellency enhances preferential water flow through continuous macropores under unsaturated conditions.

As mentioned above, part of the water applied by the rainfall simulator formed into overland flow. However, no sign of continuous overland flow (e.g. litter transportation or rill erosion) was evident on the ground surface of the slope. In a review of the hydrogeomorphological effect of water repellency, Doerr *et al.* (2000) suggested that

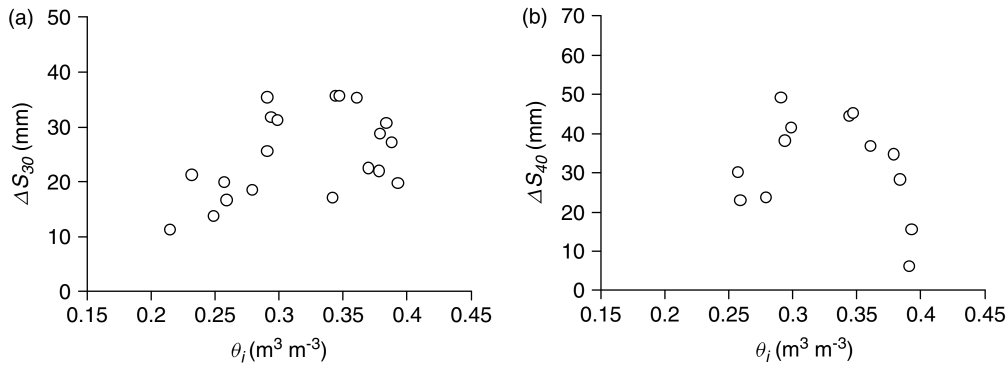

Figure 8. Relationships between the initial water content at a depth of 10 cm (θ_i) and increases in water storage within a depth of 60 cm during rainfall events for cumulative rainfalls of (a) 30 mm (ΔS_{30}) and (b) 40 mm (ΔS_{40})

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repellency induced overland flows on hillslopes are not necessarily continuous over entire slopes. Rather, the frequency of gaps such as cracks or root holes in the soil surface determines whether overland flow is continuous or is limited to a local area. Without gaps, rainwater will run downslope as continuous Hortonian overland flow. However, gaps (macropores) will intercept overland flow and transform it into vertical preferential flow. The observed water flow in this study corresponded to the latter pattern. Shakesby *et al*. (2000) also suggested water entry through macropores as the reason why continuous overland flow does not occur in burned and unburned pine forests in Portugal, even though soils in that area exhibit strong water repellency. Our data provide field evidence that repellency induced overland flow is intercepted by macropores and infiltrates into the soil as bypass flow through the macropores.

Effect of antecedent moisture conditions on flow patterns and water storage abilities

Increases in water storage during natural rainfall events were restricted under dry antecedent soil moisture conditions (Figures 7 and 8). Canopy interception may have affected the loss of rainwater and caused the considerable variation in the data. However, the observed differences between water storage increase and cumulative rainfall were much larger than expected for canopy interception by Japanese cypress (i.e. approximately 20% of the total rainfall in an event; Hattori and Chikaarashi, 1988). The tendency for water storage ability to decrease with dryness cannot be explained by the usual flow theory in which normal soil wettability is assumed, but can be interpreted by considering soil water repellency under dry conditions.

Our observations and measurements suggest the following process, illustrated in Figure 9. Under moderate antecedent moisture conditions (Figure 9a), rainwater entered the soil at many locations on the soil surface. Water would have then flowed diffusively, and most of the soil matrix could have participated in water flow and storage. Under dry moisture conditions (Figure 9b), on the other hand, rainwater flowed preferentially through spatially limited macropores and bypassed the matrix of the upper soil because of water repellency. In the deeper soil, the rate of water application through macropores was thought to be very high and to exceed the rate of water absorption by the soil matrix. A large part of the rainwater then probably continued to flow through macropores and bypassed a considerable portion of the matrix in the deeper soil apart from the preferential pathways. Consequently, a large part of the empty pore space could not participate in water storage, reducing the ability of the soil to store water at the solum scale.

Dekker and Ritsema (1996) reanalyzed experimental data collected by Bouma *et al*. (1981), which were used to examine the relationship between soil moisture and preferential flow for an undisturbed column of water repellent grassland soil. They found that water absorption by the soil was restricted not only under wet soil moisture conditions but also under dry conditions. They concluded that the dry end cases were caused by repellency induced preferential flows. Our results confirmed that the same restricted increases in soil water storage occur in a forest soil under field conditions.

When a well structured soil receives intense rainfall, the water can reach deeper parts of the soil through relatively large pores before the water spreads throughout the smaller pores. This flow condition is referred to as "physical nonequilibrium" (Ross and Smettem, 2000) and is similar to the concept of mobile–immobile water in solute transport (Gaudet *et al*., 1977). Under dry conditions, incomplete water storage would have occurred in the solum without equilibrium. Under moderate moisture conditions, water entry into the soil at the soil surface was not limited, allowing approximate equilibrium of the horizontal water distribution at each depth in the soil profile to be achieved. Therefore, water storage under moderate moisture conditions can be modeled using hydraulic properties measured under static or steady state methods.

Possible implications for hydrology at larger scales

In the studied catchment, overland flow did not seem to be a major process of water transport at the slope or catchment scales; rather, the occurrence of overland flow was supposed to be limited to a local scale due to frequent water capture by macropores. This circumstance suggests

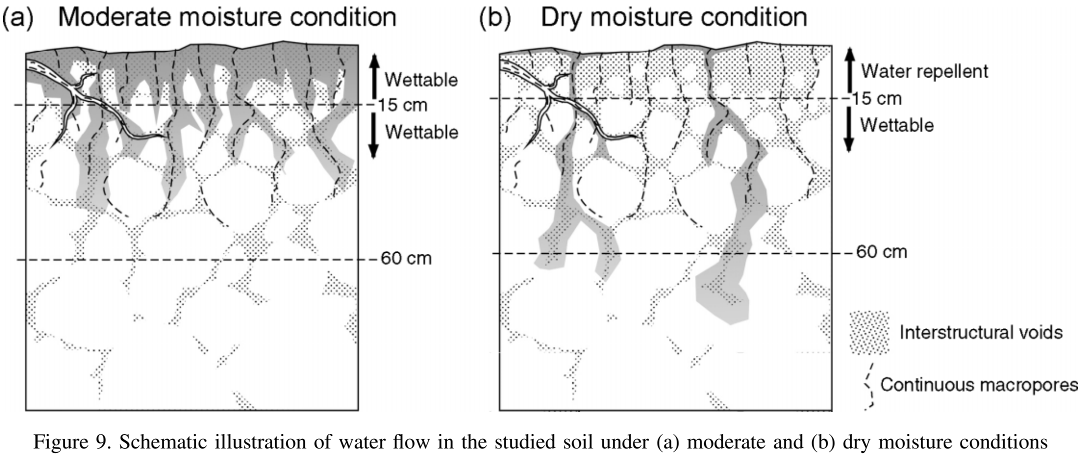

Figure 9. Schematic illustration of water flow in the studied soil under (a) moderate and (b) dry moisture conditions

that vertical preferential infiltration through macropores was the major process.

A large number of forested catchments may exist in which the soil hydrological processes are similarly affected by soil water repellency. However, the impact of soil water repellency may vary widely depending on the status of the surface soil, which is susceptible to change. Thus, there may be some unburned catchment in which overland flow is the dominant water transport process.

The findings of the present study can be applied in future research to examine whether soil water repellency affects the hydrological behavior of slopes or catchments. Although previous studies have intensively studied runoff generation on burned slopes or catchments (Krammes and DeBano, 1965; Scott, 1997; DeBano, 2000), cases of unburned forested catchments have rarely been reported (Burch *et al*., 1989; Doerr *et al*., 2003). Further research to clarify the links among soil water repellency, plot scale water movement, and catchment runoff for a variety of forested watersheds is needed.

CONCLUSIONS

We investigated water storage in a water repellent soil during rainfall events on a slope planted with Japanese cypress. The findings of this study are summarized as follows.

1. Soil water repellency occurred at below a threshold water content of approximately 0·29 m^3 m^{-3}. This value is regarded as the critical water content of this soil for water repellency. Water repellency peaked at a water content of around 0·22 m^3 m^{-3}.
2. The propagation of changes in water content to the deeper soil during rainfall events was faster under dry antecedent moisture conditions than under moderate moisture conditions. Repellency induced preferential flow through continuous macropores was thought to be responsible.
3. The occurrence of repellency induced preferential flow was confirmed by the results of rainfall simulations; the flow pattern was discontinuous and scattered under a dry moisture condition and more diffuse under a moderate condition.
4. Increases in water storage within the upper 60 cm of the soil during rainfall events were restricted under dry antecedent moisture conditions, despite the low wetness of the soil matrix. Owing to water repellency, rainwater bypassed the water repellent upper soil and a considerable part of the wettable subsoil, reflecting a physical non-equilibrium in soil water storage.

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